SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:20**

Annexure A

CLASS TEST

Criteria	Understanding of Task *(2)*	Procedure & Execution *(2.5)*	Observation & Result *(2)*	Viva Voce / Conceptual Response *(2)*	Time Management & Presentation *(1.5)*	*Total(10)*
Weightage	*20%*	25%	20%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Braneesh.A (1922571006) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Q1. **Total Marks: 10x2 =20**

a) Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

b) What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q2.

a)Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

b). What is covariance analysis? List the two covariance formals[L2, L3]

Q3.

a) Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

b) Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q4..

a) Find covariance for following data set x = {4,5,6,7,9}, y = {8,3,7,5,6}

b) Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q5

a). A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

b). Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, l5]

Data warehousing and Data mining

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CO2-AT-2 Class-Test

87/100

(1) Data mining is the process of discovering useful patterns, relationship & knowledge from large datasets using statistical, machine learning & database techniques

Two Main tasks

(1) classification

(2) Clustering

KDD: knowledge discovery in (database) process

(1) Data selection

(2) Data cleaning

(3) Data integration

(4) Data Transformation

Role of Data pre-preprocessing

+ Removes error & inconsistencies

+ Handles missing values

+ Reduced Noise

+ Improves data quality

(b) data cleaning: Is the process of detecting & correcting inaccurate, incomplete or inconsistent data

Two data Quality Issues & Handling Missing values

+ Missing values
+ Noisy data

+ Ignore records
+ Fill with mean, median or mode

Example

(2)

(a) Data integration

+ Data integration combines data from different sources into one consistent dataset

Challenges

+ Data redundancy
+ Data inconsistency

Handling redundancy

+ Remove duplicate records
+ use correlation Analysis

Handling Inconsistency

+ standardize formats

(b) Covariance Analysis

+ ~~for~~ covariance measures how two variables change together
+ Positive covariance → Increase together
+ Negative covariance → Move in opposite directions.

a:)

$A = (2, 3, 5, 4, 6)$

$B = (5, 8, 10, 11, 14)$

Sample covariance $= 5$

* Since covariance is positive both stock prices generally rise or fall together.

(b) Architecture of data mining system

Data source → Database | Data warehouse → Data preprocessing → Data mining engine → Pattern evaluation → knowledge presentation → user

Components

- Data source collect data
- Data warehouse stores data
- Preprocessing cleans data
- Mining engine discovers pattern
- pattern evaluation selects useful results

4 (a)

$X = \{4, 5, 6, 7, 9\}$

$Y = \{8, 7, 7, 5, 6\}$

Sampe covariance = -0.45

* The negative covariance indicates a weak negative relationship between x & y

(b) Steps

1. Data cleaning
2. Data Integration
3. Data Transformation
4. Data reduction
5. Data Discretization

Data-Preprocessing

Pipeline

* Remove duplicates
* Fill missing value
* Remove outliers
* Standardize formats.

2

(a) using equal-frequency (equal-depth) binning with 3 bins:

- Bin 1 Mean = 18
- Bin 2 Mean ≅ 28.11
- Bin 3 Mean ≅ 43.75

These mean values replace the original values in each bin to smooth the data.

(b) Data Discretization

- converts continuous data into interval (bins)

Importance

- Reduce noise
- simplifies data
- Improves mining performance

{Data Discretization}

Equal-width	Equal-frequency
Equal Interval size	Equal number of records
simple	Better for skewed data
Fixed width	variable width